

Parameter Estimation in a Continuously Monitored Non-Markovian Quantum System

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Continuous monitoring of quantum systems is a popular method in understanding quantum systems. Specifically, it is used increasingly as a non-demolition method for quantum metrology. In this setting, the acquired seemingly noisy data via the continuous monitoring can be used to precisely pinpoint unknown parameters of the system. It is notorious to extend the theory that connects the data to the parameter beyond Markovian dynamics. The difficulty arises from the fact that non-Markovian dynamics cannot be expressed as divisible maps. Here, we propose a method, based on the reaction coordinate mapping, to extend these techniques to systems with non-Markovian dynamics, specifically linear systems undergoing Gaussian (e.g., homodyne) continuous measurements. We showcase how our method works through an example of thermometry of a bosonic bath.

Introduction.—The theory of quantum metrology or parameter estimation is dedicated to advising measurement schemes to extract unknown parameters of quantum systems. Among its main aims are finding the ultimate precision bounds given a certain set of resources—like time, energy, or number of probes—and advising how to reach those bounds in practice.

The challenge: non-divisibility.—Any attempt to use the data collected from a continuously monitored system with a non-Markovian dynamics will have to face the non-divisibility challenge.

A go around solution: Reaction coordinate mapping.—To overcome this issue, we use a fairly well-established technique in the open quantum systems literature, namely that of the reaction coordinate mapping. Before a technical description, let us summarize the intuition behind it. As depicted in Fig. 1, for a system coupled to a linear bath one can always find an equivalent description in which the system interacts (potentially strongly) with one or more virtual bath degrees of freedom, often termed the *reaction coordinate(s)*^{cite somewhere appendix A}. In turn, the reaction coordinate(s) are weakly coupled to the *residual bath*. Thus, the reaction coordinate(s) not only directly exchange information with the system, but also mediate further information exchange between the system and the residual bath. Thanks to the weak interaction with the residual bath, one can describe the dynamics of the total system-reaction coordinate (S-RC) unit with a Markovian master equation. The continuous measurement operator acting originally on the system is now seen as a measurement on the S-RC, which can be described using standard techniques in continuous monitoring. An illustrative approach to describe the conditional evolution of a typical quantum system is to break down the evolution of the system into

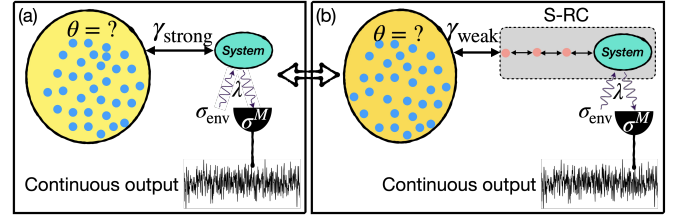


Figure 1. (a) The continuously monitored system, has a non-Markovian dynamics due to the strong coupling with the bath. The output signal is not straightforwardly interpretable, hence it cannot be immediately used to infer unknown bath parameters. (b) The reaction coordinate mapping resolves the non-Markovianity issue, hence recasting the problem as a Markovian one, where the output signal can be analyzed efficiently. Our proposal works for any Gaussian measurement, represented by covariance matrix σ^M (e.g., Homodyne detection), where the detector is placed in an environment with a Gaussian state represented by its covariance matrix σ_{env} .

consecutive action of infinitesimal maps. Namely, given that the initial state of the system is $\rho(0)$ we have

$$\rho_S(t|\vec{x}) = M_{x_n}(\lambda\delta t) \circ \mathcal{E}_{\delta t} \dots M_{x_1}(\lambda\delta t) \circ \mathcal{E}_{\delta t} \circ \rho_S(0). \quad (1)$$

Here, $\vec{x} = \{x_n, \dots, x_1\}$ is the sequence of observed outcomes, and $\delta t = t/n$. At the limit of $\delta t \rightarrow 0$ we will have a continuous signal. The operator $\mathcal{E}_{\delta t}$ is a completely positive and trace-preserving (CPTP) map representing an infinitesimal evolution of the system, undisturbed by the measurement. The maps $M_{x_k}(\lambda\delta t)$ take an input state into an output state conditioned on observing the outcome x_k . Here, we take it to preserve the norm of the state, i.e., $\text{Tr}[M_{x_k} \circ \rho] = \text{Tr}[\rho]$, $\forall \rho$. However, it is also common to take them to reduce the norm to the probability of observing the specific outcome. Furthermore, in this notation, λ represents the strength of the measurement. Crucially, the conditional evolution (1) holds true only if the dynamics is divisible, which is a notion

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of Markovianity. The issue with non-Markovian (non-divisible) dynamics is that, the dynamics operator $\mathcal{E}_{\delta t}$ is time-dependent, and thus the conditional dynamics does not admit Eq. (1).

We resolve this issue by using the reaction coordinate mapping. As illustrated in Fig. 1, and further discussed in details in the example below, after the reaction coordinate mapping—see Appendix A for more details on the RC mapping—the system-reaction coordinate unit (S-RC) undergoes a Markovian (divisible) dynamics, which admits Eq. (1). By denoting the initial state of the unit with $\rho(0) = \rho_S(0) \otimes \rho_{RC}(0)$, we have

$$\rho(t|\vec{x}) = \tilde{M}_{x_n}(\lambda\delta t) \circ \tilde{\mathcal{E}}_{\delta t} \dots \tilde{M}_{x_1}(\lambda\delta t) \circ \tilde{\mathcal{E}}_{\delta t} \circ \rho(0), \quad (2)$$

with $\tilde{M}_{x_k} = M_{x_k} \otimes I_{RC}$, and $\tilde{\mathcal{E}}_{\delta t}$ describes the dissipative dynamics of the S-RC. From here, it is straightforward to carry out the task of estimating the parameter θ e.g., in the Bayesian formalism. Recall that, in the Bayesian formalism, upon starting from an

Monitoring a non-Markovian system with a continuous Gaussian measurement.—Let us now address how one can perform parameter estimation by continuous monitoring of a linear bosonic system. Let us first remark that linear Bosonic systems with Gaussian states are very effective to describe. The reaction coordinate mapping preserves Gaussianity. However, the continuous measurement itself can create non-Gaussianity in the system, making its effective description inefficient. Nonetheless, Gaussian measurements such as homodyne currents are an important family of measurements—often performed in the quantum optical platforms—that preserve Gaussianity of the system, and hence will be our focus here.

Describing the dynamics of Gaussian CV systems is more efficient in terms of their first moments (displacement vector) and the second moments (covariance matrix). Take $R = [x_S, p_S, x_{RC}^1, p_{RC}^1, \dots, x_{RC}^m, p_{RC}^m]^T$, being the vector of quadrature operators of the S-RC, where we take the reaction coordinate modes to be $m \geq 1$. The quadratures respect canonical relations $[R_j, R_k] = i\hbar\Omega_{jk}$, with $\Omega = \oplus_{l=1}^{2m+2} \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Throughout we set $\hbar = 1$. Then, $d \in \mathbb{R}^{2m+2}$ and $\sigma \in \mathbb{R}^{2m+2} \times \mathbb{R}^{2m+2}$ are defined as

$$\begin{aligned} d_j &:= \text{Tr}[\rho R_j], \\ \sigma_{jk} &:= \frac{1}{2} \text{Tr}[\rho \{R_j, R_k\}] - d_j d_k. \end{aligned} \quad (3)$$

The covariance matrix respects the Heisenberg uncertainty relation $\sigma \pm i\Omega/2 \geq 0$. Without the measurement outcome considered, the (Gaussian) Markovian dynamics of the S-RC can always be described as follows

$$\begin{aligned} d(\theta) &\mapsto d(\theta) + \delta t A(\theta) d(\theta) \\ \sigma(\theta) &\mapsto \sigma(\theta) + \delta t [A(\theta)\sigma(\theta) + \sigma(\theta)A^T(\theta) + D(\theta)], \end{aligned} \quad (4)$$

where $A(\theta) = A_1(\theta) + A_2$ with $A_k \in \mathbb{R}^{2m+2} \times \mathbb{R}^{2m+2}$ being the drift part of the dynamics corresponding to the sample bath ($A_1(\theta)$) and the detector environment

(A_2). Furthermore, $D(\theta) = D_1(\theta) + D_2$ with $D_k^T = D_k \in \mathbb{R}^{2m+2} \times \mathbb{R}^{2m+2}$ is the diffusion part for the sample bath ($D_1(\theta)$) and the detector environment (D_2). While the specific forms of $A_1(\theta)$ and $D_1(\theta)$ depend on the details of the original system-bath Hamiltonian, for A_2 and D_2 they are fairly simpler. We will take them to be $A_2 = -\lambda_2 I_S \oplus 0_{RC}$ and $D_2 = \lambda \sigma_{\text{env}} \oplus 0_{RC}$, where σ_{env} is the covariance matrix of the environment modes where the detector is located. Often this is the vacuum represented by $\sigma_{\text{env}} = I/2$ and $d_{\text{env}} = 0$. However, we can leave it to be a general Gaussian state with zero mean $d_{\text{env}} = 0$.

Incorporating the measurement outcome requires to determine what specific Gaussian measurement we are performing. This can be represented by a physical covariance matrix $\sigma^M \in \mathbb{R}^2 \times \mathbb{R}^2$. Common choices are Homodyne detection of say x_S quadrature corresponding to $\sigma^M = \lim_{r \rightarrow 0} \text{diag}[r, 1/r]/2$, or Heterodyne detection corresponding to $\sigma^M = I/2$. Given the S-RC state of the general form

$$\begin{aligned} d(\theta) &= [d_S(\theta), d_{RC}(\theta)], \\ \sigma(\theta) &= \begin{bmatrix} \sigma_S(\theta) & C(\theta) \\ C^T(\theta) & \sigma_{RC}(\theta) \end{bmatrix}, \end{aligned} \quad (5)$$

the Gaussian measurement will output $W \in \mathbb{R}^2$ according to the normal distribution

$$p(W|\theta) = \mathcal{N}[\xi(\theta), V(\theta)], \quad (6)$$

with the mean $\xi(\theta) = \sqrt{\lambda\delta t} d_S(\theta)$ and the variance $V(\theta) = \sigma_{\text{env}} + \lambda\delta t(\sigma_S(\theta) - \sigma_{\text{env}}) + \sigma^M$. If in $p(W|\theta)$ we only keep terms of order δt and less, then we have $V(\theta) \approx \sigma_{\text{env}} + \sigma^M = V$. Conditioned on observing this outcome, the S-RC will then evolves (update) to

$$\begin{aligned} d(\theta|W) &= d(\theta) + \delta t A(\theta) d(\theta) + \sqrt{\lambda\delta t} \begin{bmatrix} \sigma_S(\theta) - \sigma_{\text{env}} \\ C^T(\theta) \end{bmatrix} \\ &\quad \times V^{-1}(W - \sqrt{\lambda\delta t} d_S(\theta)), \\ \sigma(\theta|W) &= \sigma(\theta) + \delta t [A(\theta)\sigma(\theta) + \sigma(\theta)A^T(\theta) + D(\theta)] \\ &\quad - \lambda\delta t \begin{bmatrix} \sigma_S(\theta) - \sigma_{\text{env}} \\ C^T(\theta) \end{bmatrix} V^{-1} [\sigma_S(\theta) - \sigma_{\text{env}}, C(\theta)]. \end{aligned} \quad (7)$$

Note that the covariance matrix actually does not depend on the specific observed outcome, and its W dependence may be dropped. The Eqs. (7) provide a stochastic iterative algorithm to describe the time evolution of the S-RC. While the system alone cannot be described by a Markovian dynamics, these equations allow the dynamics of the S-RC to be described as a stochastic Markovian dynamics. In principle, the reaction coordinates capture the entire memory of the environment about the system, hence putting them in the same unit as the system enables us to efficiently describe the conditional dynamics.

Example: monitoring a non-Markovian Brownian particle.—To illustrate our proposed techniques, we consider the pedagogical quantum Brownian motion problem, with the total Hamiltonian

$$\hat{H} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB}. \quad (8)$$

Here, $\hat{H}_S = \frac{1}{2} [p_S^2 + (\Omega_S^2 + \delta\Omega_S^2)x_S^2]$ describes a harmonic oscillator (the Brownian particle) with frequency Ω_S which is [strongly] coupled to a bosonic bath at temperature T with Hamiltonian $\hat{H}_B = \frac{1}{2} \sum_k (p_k^2 + \omega_k^2 x_k^2)$. The interaction between the system and bath is modeled by a position-position interaction like $\hat{H}_{SB} = -x_S \sum_k c_k x_k$. The Brownian particle is continuously monitored to infer some property of the bath, say its temperature. Here, $\delta\Omega_S^2$ is the bath-induced frequency renormalization term. Throughout the text we assume unit masses for both system and bath modes. To proceed further, we define the bath spectral density as,

$$\mathcal{J}(\omega) := \frac{\pi}{2} \sum_k \frac{c_k^2}{\omega_k} \delta(\omega - \omega_k). \quad (9)$$

The interaction Hamiltonian introduces a renormalization frequency shift to the system,

$$\delta\Omega_S^2 := \sum_k \frac{c_k^2}{\omega_k^2} = \frac{2}{\pi} \int_0^\infty d\omega \frac{\mathcal{J}(\omega)}{\omega}. \quad (10)$$

In what follows, we take a Lorentzian profile for the spectral density

$$\mathcal{J}(\omega) = \frac{\gamma g^2 \omega}{\gamma^2 \omega^2 + (\omega^2 - \Omega_{RC}^2)^2}. \quad (11)$$

This choice describes a narrow-band environment peaked around Ω_{RC} , with γ controlling the bath memory time and g the system-bath coupling strength; it is precisely the class of structured reservoirs for which the RC mapping is expected to be especially effective [1, 2].

Given an initial uncorrelated state of the system-bath, where the system is any Gaussian state and the bath is at thermal equilibrium, we can exactly solve the dynamics of the Brownian particle by means of the quantum Langevin equation—see Appendix B 1. However, the conditional dynamics subject to continuous monitoring requires further treatment of the model, namely using the aforementioned reaction coordinate mapping.

The reaction coordinate mapping.—We now showcase how to define the reaction coordinate and write a [global] master equation for the S-RC unit, starting from the original Hamiltonian (8). The reaction coordinate is defined as a collective bath coordinate, namely,

$$X_{RC} := \frac{1}{g} \sum_k c_k x_k, \quad (12)$$

with $g^2 = \sum_k c_k^2$. This allows us to rewrite the Hamiltonian as a function of the RC position and momentum operators:

$$\begin{aligned} \hat{H} = & \hat{H}_S + \frac{1}{2} (P_{RC}^2 + \Omega_{RC}^2 X_{RC}^2) - g x_S X_{RC} \\ & - X_{RC} \sum_{k>1} C_k X_k + \frac{1}{2} \sum_{k>1} (P_k^2 + \Omega_k^2 X_k^2), \end{aligned} \quad (13)$$

where now the sum involving bath operators starts at $k > 1$, as we have extracted one mode from the bath. Here, the coefficients C_k and residual mode frequencies Ω_k are set by the canonical transformation defining the RC mapping. The mapping determines both the RC frequency, $\Omega_{RC}^2 = \frac{1}{g^2} \sum_k c_k^2 \omega_k^2$, and the residual bath spectral density, $\mathcal{J}_{RC}(\omega) = \gamma \omega e^{-|\omega|/\Lambda}$, which is Ohmic in the limit of $\Lambda \rightarrow \infty$, with Λ being the cut-off frequency.

Global GKLS master equation for the S-RC.—There are scenarios where the original dynamics is non-Markovian, but the S-RC can be effectively described by a Markovian master equation. If the S-RC is still not behaving in a Markovian manner, one may need to further include more reaction coordinates, as discussed before. Here, we work with a regime of parameters where only one reaction coordinate mode is sufficient for a Markovian S-RC description—see Appendix B 2 for further details on the derivation. We also compare this approach with a local GKLS treatment, detailed in Appendix B 3, and with the exact Langevin solution in Appendix B 4.

To derive the master equation for the S-RC, we note that generally speaking the interaction between the system and the reaction coordinate mode will be strong, which is inherited from the strong coupling of the system to the original bath. As such, it is necessary to take a global master equation approach. One can write down the dissipative dynamics of the S-RC in the form of (4), with

$$\begin{aligned} A_1(\theta) = \\ D_1(\theta) = \end{aligned} \quad (14)$$

If we have enough space, we will write down everything, but that means a big sequence of equations tracing back A_1 and D_1 to the parameters defined so far. Otherwise, this can be removed, only refer to the appendix and mention that we have obtained them analytically, and their dependence on the parameter θ are clearly stated.

As for the specific Gaussian measurement, let's take Homodyne detection of the system's x_S quadrature. Furthermore, we assume the detector environment is at zero temperature.

The Bayesian estimation.—Our formalism provides all the key ingredients for the Bayesian estimation of any parameter of the bath. Starting from a prior knowledge $p(\theta)$, and an initial system state, Eq. (6) gives $p(W|\theta)$ the likelihood of the outcomes. This can be used to build up the posterior distribution thanks to the Bayes' update rule $p(\theta|W) = p(W|\theta)p(\theta)/p(W)$ with $p(W) = \int d\theta p(W|\theta)p(\theta)$. Furthermore, Eqs. (7) provide the instructions for updating the (conditional) input state, which will be used as the arguments of the likelihood distribution for the next outcome and so on. At any given time, the posterior distribution can be used to assign an estimate to the unknown parameter.

To proceed further, let us take the specific example of thermometry, i.e., T is unknown. In Bayesian thermometry it is common practice to define the mean square

logarithmic error (MSLE) as a figure of merit, defined as

$$\text{MSLE} = . \quad (15)$$

For this choice, the optimal estimator is analytically proven to be

$$T^*(W) = . \quad (16)$$

In Fig. ?? we demonstrate a Monte Carlo simulation of such an estimation task. Panel (a) shows the noisy outcome signal, while panel (b) shows the posterior distribution over time, overlaid with the estimate temperature.

Lastly, panel (c) shows the logarithmic error of the estimator, which overall drops with time.

Discussion.—Continuously monitoring a non-Markovian system generally can be challenging due to the non-divisibility of the dynamics. Our go-around solution which uses the reaction coordinate mapping is a viable solution. As we demonstrated, it can be used to interpret an otherwise complex signal and hence infer unknown parameters of the bath.

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Appendix A: On the reaction coordinate mapping

In this section, we derive all the details regarding the reaction coordinate mapping used in the main text—namely, how the Hamiltonian and the spectral density are transformed after applying the mapping.

1. The RC Hamiltonian

Let us start by computing the Hamiltonian after the RC mapping. We mainly follow the derivation introduced in Refs. [1, 2].

As introduced in Eq. (12), the reaction coordinate is defined by extracting a collective mode from the bath. Hence, we map the N modes of the original bath into a RC plus $N - 1$ residual bath modes. This can be implemented by an orthogonal transformation of the form

$$X_i = \sum_j O_{ij} x_j, \quad (\text{A1})$$

$$P_i = \sum_j O_{ij} p_j. \quad (\text{A2})$$

Note that the transformation needs to be the same for both operators to preserve the canonical commutation relation:

$$[X_i, P_j] = \sum_{m,l} [O_{im} x_m, O_{jl} p_l] = \sum_{m,l} O_{im} O_{jl} \underbrace{[x_m, p_l]}_{i\delta_{ml}} = i \sum_{m,l} O_{im} O_{jl} \delta_{ml} = i \sum_m O_{im} O_{jm} = i\delta_{ij}, \quad (\text{A3})$$

as O is an orthogonal matrix (i.e., $O^T = O^{-1}$). If the transformation is such that the residual baths also have a normal form, the following relation holds:

$$\sum_k O_{ik} \omega_k^2 O_{jk} = \delta_{ij} \Omega_i^2, \quad (\text{A4})$$

with $i, j > 1$, since the first row of the transformation is fixed by Eq. (12). Using these insights, we can rewrite the Hamiltonian as a function of the RC operators:

$$\hat{H} = \hat{H}_S + \frac{1}{2}(P_{\text{RC}}^2 + \Omega_{\text{RC}}^2 X_{\text{RC}}^2) - g x_S X_{\text{RC}} - X_{\text{RC}} \sum_{k>1} C_k X_k + \frac{1}{2} \sum_{k>1} (P_k^2 + \Omega_k^2 X_k^2), \quad (\text{A5})$$

with $C_k = -\sum_m \omega_m^2 O_{m1} O_{mk}$. Observing that $O_{1k} = c_k/g$ we can derive a constraint for g from the new canonical commutation relations for the RC operators:

$$[X_{\text{RC}}, P_{\text{RC}}] = \sum_{m,l} [O_{1m} x_m, O_{1l} p_l] = \sum_{m,l} O_{1m} O_{1l} [x_m, p_l] = i \sum_{m,l} O_{1m} O_{1l} \delta_{ml} = i \sum_m O_{1m}^2 \quad (\text{A6})$$

$$= \frac{i}{g^2} \sum_m c_m^2 = i \Rightarrow g^2 = \sum_k c_k^2. \quad (\text{A7})$$

Now, let us relate all these quantities to the original spectral density, so that we can know all the parameters that appear in the transformation by just knowing the spectral density. To do that, note from Eq. (A4) that the RC frequency fulfills

$$\Omega_{\text{RC}}^2 = \sum_k \omega_k^2 O_{1k}^2 = \frac{1}{g^2} \sum_k c_k^2 \omega_k^2. \quad (\text{A8})$$

If we use the spectral density in Eq. (9), the following relations hold:

$$\delta\Omega_S^2 := \sum_k \frac{c_k^2}{\omega_k^2} = \frac{2}{\pi} \int_0^\infty d\omega \frac{\mathcal{J}(\omega)}{\omega}, \quad (\text{A9})$$

$$g^2 = \frac{2}{\pi} \int_0^\infty d\omega \omega \mathcal{J}(\omega), \quad (\text{A10})$$

$$\Omega_{\text{RC}}^2 = \frac{2}{\pi g^2} \int_0^\infty d\omega \omega^3 \mathcal{J}(\omega). \quad (\text{A11})$$

Now, it is left to relate $\mathcal{J}(\omega)$ with the spectral density of the RC, $\mathcal{J}_{\text{RC}}(\omega)$. We give an analogous proof to the ones in the literature [2–5], which is based on the fact that the Fourier space propagator in both coordinate systems must be the same. The obtained expression—see Appendix A 2—is given by:

$$\mathcal{J}_{\text{RC}}(\omega) = \frac{g^2 \mathcal{J}(\omega)}{|W^+(\omega)|^2}, \quad (\text{A12})$$

where $W^+(\omega)$ is the Cauchy transform of the original spectral density, which is given by

$$W(z) = \frac{1}{\pi} \int_{-\infty}^{+\infty} d\omega \frac{\mathcal{J}(\omega)}{\omega - z} \quad (\text{A13})$$

with

$$W^+(\omega) := \lim_{\epsilon \searrow 0} W(\omega + i\epsilon), \quad \omega \in \mathbb{R}, \quad (\text{A14})$$

where the limit emphasizes that we approach ω from the upper half of the complex plane. Therefore, the RC spectral density takes the form

$$\mathcal{J}_{\text{RC}}(\omega) = \frac{\pi}{2} \sum_k \frac{C_k^2}{\Omega_k} \delta(\omega - \Omega_k), \quad (\text{A15})$$

and the following relation also holds:

$$\frac{g^2}{\delta\Omega_{\text{S}}^2} = \Omega_{\text{RC}}^2 - \delta\Omega_{\text{RC}}^2, \quad (\text{A16})$$

with

$$\delta\Omega_{\text{RC}}^2 := \sum_k \frac{C_k^2}{\Omega_k^2}. \quad (\text{A17})$$

Using Eq. (A15) we can rewrite the total Hamiltonian as

$$H = \frac{1}{2}(p_{\text{S}}^2 + \Omega_{\text{S}}^2 x_{\text{S}}^2) + \frac{1}{2} \left[P_{\text{RC}}^2 + \frac{g^2}{\delta\Omega_{\text{S}}^2} \left(X_{\text{RC}} - \frac{\delta\Omega_{\text{S}}^2}{g} x_{\text{S}} \right)^2 \right] + \frac{1}{2} \sum_k \left[P_k^2 + \Omega_k^2 \left(X_k - \frac{C_k}{\Omega_k^2} X_{\text{RC}} \right)^2 \right] \quad (\text{A18})$$

which is in Brownian form and reproduces (up to the shift in the system Hamiltonian) the RC Hamiltonian in Eq. (13).

2. The RC spectral density

From the Hamiltonian in Eq. (8), the Hamilton equations of motion can be shown to be

$$\ddot{x}_{\text{S}}(t) = -\Omega_{\text{S}}^2 x_{\text{S}}(t) - x_{\text{S}}(t) \sum_k \frac{c_k^2}{\omega_k^2} + \sum_k x_k(t) c_k, \quad (\text{A19})$$

$$\ddot{x}_k(t) = -\omega_k^2 x_k(t) + c_k x_{\text{S}}(t). \quad (\text{A20})$$

Let us now Fourier transform both equations such that $\tilde{f}(z) := \int_{-\infty}^{+\infty} dt f(t) e^{izt}$:

$$-z^2 \tilde{x}_{\text{S}}(z) = -\Omega_{\text{S}}^2 \tilde{x}_{\text{S}}(z) - \tilde{x}_{\text{S}}(z) \sum_k \frac{c_k^2}{\omega_k^2} + \sum_k \tilde{x}_k(z) c_k, \quad (\text{A21})$$

$$-z^2 \tilde{x}_k(z) = -\omega_k^2 \tilde{x}_k(z) + c_k \tilde{x}_{\text{S}}(z). \quad (\text{A22})$$

Substituting the second equation into the first one, we get:

$$-\Omega_{\text{S}}^2 \tilde{x}_{\text{S}}(z) = \tilde{\mathcal{L}}_{\text{S}}(z) \tilde{x}_{\text{S}}(z), \quad \tilde{\mathcal{L}}_{\text{S}}(z) := -z^2 + \sum_k \frac{c_k^2}{\omega_k^2} - \sum_k \frac{c_k^2}{\omega_k^2 - z^2}. \quad (\text{A23})$$

Now, we have to do the same trick with the Hamiltonian in Eq. (13). In this case, we have three equations of motion, given by

$$\ddot{x}_S(t) = -\Omega_S^2 x_S(t) - \delta\Omega_S^2 x_S(t) + gX_{RC}(t), \quad (\text{A24})$$

$$\ddot{X}_{RC}(t) = -\Omega_{RC}^2 X_{RC}(t) + gx_S(t) + \sum_k C_k X_k(t), \quad (\text{A25})$$

$$\ddot{X}_k(t) = -\Omega_k^2 X_k(t) + C_k X_{RC}(t). \quad (\text{A26})$$

Fourier-transforming, we get:

$$-z^2 \tilde{x}_S(z) = -\Omega_S^2 \tilde{x}_S(z) - \delta\Omega_S^2 \tilde{x}_S(z) + g\tilde{X}_{RC}(z), \quad (\text{A27})$$

$$-z^2 \tilde{X}_{RC}(z) = -\Omega_{RC}^2 \tilde{X}_{RC}(z) + g\tilde{x}_S(z) + \sum_k C_k \tilde{X}_k(z), \quad (\text{A28})$$

$$-z^2 \tilde{X}_k(z) = -\Omega_k^2 \tilde{X}_k(z) + C_k \tilde{X}_{RC}(z), \quad (\text{A29})$$

from where we can obtain

$$-\Omega_S^2 \tilde{x}_S(z) = \tilde{\mathcal{L}}(z) \tilde{x}_S(z), \quad \tilde{\mathcal{L}}(z) := -z^2 + \delta\Omega_S^2 - \frac{g^2}{-z^2 + \Omega_{RC}^2 - \sum_k \frac{C_k^2}{\Omega_k^2 - z^2}}. \quad (\text{A30})$$

As we said, both Fourier propagators must be equal, so the following relation must hold:

$$-z^2 + \sum_k \frac{c_k^2}{\omega_k^2} - \sum_k \frac{c_k^2}{\omega_k^2 - z^2} = -z^2 + \delta\Omega_S^2 - \frac{g^2}{-z^2 + \Omega_{RC}^2 - \sum_k \frac{C_k^2}{\Omega_k^2 - z^2}} \Rightarrow W_0(z) = \frac{g^2}{-z^2 + \Omega_{RC}^2 - W_1(z)}, \quad (\text{A31})$$

where we have defined the Cauchy transform of $\mathcal{J}(\omega)$ as (same for the RC spectral density)

$$W_0(z) = \frac{1}{\pi} \int_{-\infty}^{+\infty} d\omega \frac{\mathcal{J}(\omega)}{\omega - z}, \quad (\text{A32})$$

which can be inverted to obtain

$$W_1(z) = \Omega_{RC}^2 - z^2 - \frac{g^2}{W_0(z)}. \quad (\text{A33})$$

Note that we extended $\mathcal{J}(\omega)$ to negative values of ω via $\mathcal{J}(-\omega) = -\mathcal{J}(\omega)$. From here, Eq. (A12) can be derived by taking the imaginary part of Eq. (A33) and noticing that $J(\omega) = \text{Im } W_0^+(\omega)$.

a. From Lorentzian to Ohmic

Now, let us check that Eq. (11) is recovered when imposing an Ohmic profile for the RC spectral density, that is, $\mathcal{J}_{RC}(\omega) = \gamma\omega e^{-|\omega|/\Lambda}$ in the limit of infinite cut-off frequency $\Lambda \rightarrow \infty$. Again, we have extended the RC spectral density to negative values of ω , that is, $\mathcal{J}_{RC}(-\omega) = -\mathcal{J}_{RC}(\omega)$, which justifies the absolute value in the exponential cut-off.

We start by computing $W_1^+(\omega)$, which will later give us $W_0^+(\omega)$ via Eq. (A31):

$$W_1^+(\omega) = \frac{1}{\pi} \lim_{\epsilon \searrow 0} \int_{-\infty}^{\infty} d\omega' \frac{\mathcal{J}_{RC}(\omega')}{\omega' - (\omega + i\epsilon)} = \frac{1}{\pi} \lim_{\epsilon \searrow 0} \int_{-\infty}^{\infty} dx \frac{\mathcal{J}_{RC}(\omega + x)}{x - i\epsilon} \quad (\text{A34})$$

$$= \frac{1}{\pi} \left(i\pi \int_{-\infty}^{\infty} dx \mathcal{J}_{RC}(\omega + x) \delta(x) + \mathcal{P} \int_{-\infty}^{\infty} dx \frac{\mathcal{J}_{RC}(\omega + x)}{x} \right) \quad (\text{A35})$$

$$= i\mathcal{J}_{RC}(\omega) + \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\mathcal{J}_{RC}(\omega')}{\omega' - \omega} := \Delta_1(\omega) + i\mathcal{J}_{RC}(\omega), \quad (\text{A36})$$

where we have used Sokhotski–Plemelj’s theorem to compute the limit. Let us focus on computing the principal value term for the Ohmic spectral density:

$$\Delta_1(\omega) = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\gamma \omega' e^{-|\omega'|/\Lambda}}{\omega' - \omega} = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \gamma e^{-|\omega'|/\Lambda} \left(1 + \frac{\omega}{\omega' - \omega} \right) \quad (\text{A37})$$

$$= \frac{\gamma}{\pi} \left(\mathcal{P} \int_{-\infty}^{\infty} d\omega' e^{-|\omega'|/\Lambda} + \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\omega e^{-|\omega'|/\Lambda}}{\omega' - \omega} \right) = \frac{2\gamma\Lambda}{\pi} + \frac{\gamma}{\pi} \mathcal{P} \int_{-\infty}^{\infty} d\omega' \frac{\omega e^{-|\omega'|/\Lambda}}{\omega' - \omega} \quad (\text{A38})$$

$$= \frac{2\gamma\Lambda}{\pi} + \frac{\gamma\omega}{\pi} \left(\mathcal{P} \int_0^{\infty} d\omega' \frac{e^{-\omega'/\Lambda}}{\omega' - \omega} - \mathcal{P} \int_0^{\infty} d\omega' \frac{e^{-\omega'/\Lambda}}{\omega' + \omega} \right) \quad (\text{A39})$$

$$= \frac{2\gamma\Lambda}{\pi} - \frac{\gamma\omega}{\pi} \left[e^{-\omega/\Lambda} \text{Ei} \left(\frac{\omega}{\Lambda} \right) + e^{\omega/\Lambda} \Gamma \left(0, \frac{\omega}{\Lambda} \right) \right]. \quad (\text{A40})$$

Now, let us show that the second term goes to 0 in the limit of $\Lambda \rightarrow \infty$. To do that, we expand the whole term around $\omega/\Lambda =: x \approx 0$:

$$e^{-x} \text{Ei}(x) + e^x \Gamma(0, x) \approx -2x(-1 + \gamma_E + \log x) + \mathcal{O}(x^3) \approx 0, \quad (\text{A41})$$

where γ_E is the Euler-Mascheroni constant. Note that we have used $\lim_{x \rightarrow 0} x \log x = 0$. Hence,

$$W_1^+(\omega) = \frac{2\gamma\Lambda}{\pi} + i\mathcal{J}_{\text{RC}}(\omega), \quad (\text{A42})$$

and we can compute $W_0^+(\omega)$ with Eq. (A31):

$$W_0^+(\omega) = \frac{g^2}{\Omega_{\text{RC}}^2 - \omega^2 - \frac{2\gamma\Lambda}{\pi} - i\mathcal{J}_{\text{RC}}(\omega)} = \frac{g^2 \left(\Omega_{\text{RC}}^2 - \omega^2 - \frac{2\gamma\Lambda}{\pi} + i\mathcal{J}_{\text{RC}}(\omega) \right)}{\left[\Omega_{\text{RC}}^2 - \omega^2 - \frac{2\gamma\Lambda}{\pi} \right]^2 + \mathcal{J}_{\text{RC}}^2(\omega)}. \quad (\text{A43})$$

Note that in order to extend the relation between $W_0(z)$ and $W_1(z)$ to the corresponding boundary values $W_0^+(\omega)$ and $W_1^+(\omega)$, the denominator in Eq. (A43) must not vanish. Using Eq. (A36), the denominator can be written as

$$\Omega_{\text{RC}}^2 - \omega^2 - W_1^+(\omega) = \Omega_{\text{RC}}^2 - \omega^2 - \Delta_1(\omega) - i\mathcal{J}_{\text{RC}}(\omega). \quad (\text{A44})$$

For $\omega > 0$, one has $\mathcal{J}_{\text{RC}}(\omega) > 0$, so the denominator cannot vanish. At $\omega = 0$, since $\mathcal{J}_{\text{RC}}(0) = 0$, the denominator reduces to

$$\Omega_{\text{RC}}^2 - \Delta_1(0) = \Omega_{\text{RC}}^2 - \frac{2\gamma\Lambda}{\pi}, \quad (\text{A45})$$

which is precisely the *physical* RC frequency squared in Eq. (A16). Hence, the extension to the boundary values $W_0^+(\omega)$ and $W_1^+(\omega)$ is well defined provided that the physical RC frequency does not vanish.

Finally, using $\mathcal{J}(\omega) = \text{Im } W_0^+(\omega)$, we get Eq. (11):

$$\mathcal{J}(\omega) = \frac{\gamma g^2 \omega}{\left[\Omega_{\text{RC}}^2 - \omega^2 - \frac{2\gamma\Lambda}{\pi} \right]^2 + \gamma^2 \omega^2}, \quad (\text{A46})$$

which includes $\delta\Omega_{\text{RC}}^2 = \frac{2\gamma\Lambda}{\pi}$. Indeed, by Eq. (A17) we have:

$$\delta\Omega_{\text{RC}}^2 = \sum_k \frac{C_k^2}{\Omega_k^2} = \frac{2}{\pi} \int_0^{\infty} d\omega \frac{\mathcal{J}_{\text{RC}}(\omega)}{\omega} = \frac{2}{\pi} \int_0^{\infty} d\omega \frac{\gamma \omega e^{-\omega/\Lambda}}{\omega} = \frac{2\gamma}{\pi} \int_0^{\infty} d\omega e^{-\omega/\Lambda} = \frac{2\gamma\Lambda}{\pi}. \quad (\text{A47})$$

Note that this is in agreement with Ref. [1], where the spin-boson spectral density, $\mathcal{J}(\omega)$, includes the physical reaction coordinate frequency and not Ω_{RC} , which is the case in Eq. (A47).

Appendix B: Solving the dynamics

We devote this section to present the three alternative descriptions used throughout the manuscript to solve the dynamics of the model in Eq. (8). We first give an exact solution based on the quantum Langevin equation. We then derive a global GKLS master equation for the system-RC unit and, finally, we introduce the local GKLS master equation for the reduced system that we later use as a comparison.

1. Exact solution: the quantum Langevin equation

First, we exactly solve the dynamics of the system oscillator by integrating out the bath degrees of freedom. This yields a generalized quantum Langevin equation with a non-local memory kernel, from which both the transient dynamics and the steady-state covariances can be obtained. This exact solution is later used as a benchmark for the unconditional reduced dynamics of the system oscillator. We assume an initially factorized state,

$$\rho(0) = \rho_S(0) \otimes \rho_B, \quad (\text{B1})$$

where ρ_B is a thermal state of the bath at inverse temperature $\beta = 1/T$, while $\rho_S(0)$ is an arbitrary Gaussian state of the system oscillator. We start by computing the Heisenberg equations of motion for the Hamiltonian in Eq. (8), with dots denoting time derivatives:

$$\ddot{x}_S(t) = -\Omega_S^2 x_S(t) - x_S(t) \sum_k \frac{c_k^2}{\omega_k^2} + \sum_k c_k x_k(t), \quad (\text{B2})$$

$$\ddot{x}_k(t) = -\omega_k^2 x_k(t) + c_k x_S(t). \quad (\text{B3})$$

From Eq. (B3), one can obtain a formal solution for each bath mode. To do that, we use standard differential equations techniques. First, we know that the set of solutions of the homogeneous equation is the one for the harmonic oscillator, i.e., $\{x_k^{(1)}(t), x_k^{(2)}(t)\} = \{\sin(\omega_k t), \cos(\omega_k t)\}$. The particular solution will be given by

$$x_k^p(t) = u_k^{(1)}(t)x_k^{(1)}(t) + u_k^{(2)}(t)x_k^{(2)}(t), \quad (\text{B4})$$

where $u_k^{(1)}(t)$ and $u_k^{(2)}(t)$ are defined as follows:

$$\dot{u}_k^{(1)} = \frac{1}{W[x_k^{(1)}(t), x_k^{(2)}(t)]} \det \begin{pmatrix} 0 & x_k^{(2)}(t) \\ c_k x_S(t) & \dot{x}_k^{(2)}(t) \end{pmatrix}, \quad \dot{u}_k^{(2)} = \frac{1}{W[x_k^{(1)}(t), x_k^{(2)}(t)]} \det \begin{pmatrix} x_k^{(1)}(t) & 0 \\ \dot{x}_k^{(1)}(t) & c_k x_S(t) \end{pmatrix}, \quad (\text{B5})$$

with $W[x_k^{(1)}(t), x_k^{(2)}(t)]$ being the determinant of the Wronskian matrix. To obtain $u_k^{(1)}(t)$ and $u_k^{(2)}(t)$ we just need to integrate the previous equations:

$$u_k^{(1)}(t) = -\frac{c_k}{\omega_k} \int_0^t dt' x_S(t') \sin(\omega_k t'), \quad (\text{B6})$$

$$u_k^{(2)}(t) = \frac{c_k}{\omega_k} \int_0^t dt' x_S(t') \cos(\omega_k t'). \quad (\text{B7})$$

Therefore, the final solution of Eq. (B3), including the initial conditions, is given by

$$x_k(t) = x_k(0) \cos(\omega_k t) + \frac{p_k(0)}{\omega_k} \sin(\omega_k t) + \frac{c_k}{\omega_k} \int_0^t dt' x_S(t') \sin[\omega_k(t - t')], \quad (\text{B8})$$

where we have used $\sin(A - B) = \sin A \cos B - \sin B \cos A$. Inserting Eq. (B8) into Eq. (B2) we get:

$$\ddot{x}_S(t) = -\Omega_S^2 x_S(t) - x_S(t) \sum_k \frac{c_k^2}{\omega_k^2} + \sum_k c_k \left(x_k(0) \cos(\omega_k t) + \frac{p_k(0)}{\omega_k} \sin(\omega_k t) \right) + \sum_k \frac{c_k^2}{\omega_k} \int_0^t dt' x_S(t') \sin[\omega_k(t - t')]. \quad (\text{B9})$$

By the definition of the bath spectral density in Eq. (9) and rearranging terms, we get

$$\ddot{x}_S(t) + (\Omega_S^2 + \delta\Omega_S^2)x_S(t) - \frac{2}{\pi} \int_0^\infty d\omega \mathcal{J}(\omega) \int_0^t dt' x_S(t') \sin[\omega(t - t')] = \sum_k \tilde{B}_k(t) =: \tilde{B}(t), \quad (\text{B10})$$

where we have defined

$$\tilde{B}_k(t) := c_k \left(x_k(0) \cos(\omega_k t) + \frac{p_k(0)}{\omega_k} \sin(\omega_k t) \right). \quad (\text{B11})$$

By further defining the dissipation kernel as

$$\chi(t) := \frac{2}{\pi} \int_0^\infty d\omega' \mathcal{J}(\omega') \sin(\omega' t), \quad (\text{B12})$$

we get the generalized Langevin form

$$\ddot{x}_S(t) + (\Omega_S^2 + \delta\Omega_S^2)x_S(t) - \int_0^t dt' \chi(t-t')x_S(t') = \tilde{B}(t). \quad (\text{B13})$$

Since the memory integral runs over $0 \leq t' \leq t$, the argument $t-t'$ is non-negative, and no explicit Heaviside factor is needed. The exact solution can then be written in terms of the Green function $G(t)$ as [6]

$$x_S(t) = x_S(0)\dot{G}(t) + p_S(0)G(t) + \int_0^t d\tau G(t-\tau)\tilde{B}(\tau), \quad (\text{B14})$$

where $G(t)$ and $\dot{G}(t)$ are defined in terms of the Laplace-space Green function, $\mathcal{G}(s)$. That is,

$$\mathcal{L}\{G(t)\} =: \mathcal{G}(s) = \frac{1}{s^2 + \Omega_S^2 + \delta\Omega_S^2 - \mathcal{L}\{\chi\}(s)}, \quad (\text{B15})$$

$$\mathcal{L}\{\dot{G}(t)\} = \frac{s}{s^2 + \Omega_S^2 + \delta\Omega_S^2 - \mathcal{L}\{\chi\}(s)}. \quad (\text{B16})$$

For Eq. (B14) to reproduce the right initial conditions, one must have $G(0) = 0$ and $\dot{G}(0) = 1$. In the case of the spectral density in Eq. (11), we can compute $G(t)$ and its derivative analytically, as we show in the next section.

Finally, for the p -quadrature, we just need to differentiate Eq. (B14) to obtain:

$$p_S(t) = x_S(0)\ddot{G}(t) + p_S(0)\dot{G}(t) + \int_0^t d\tau \dot{G}(t-\tau)\tilde{B}(\tau). \quad (\text{B17})$$

a. Green functions for the Lorentzian spectral density

We first redefine $g^2 = \alpha_1\alpha_2\gamma$ and $\Omega_{\text{RC}}^2 = \alpha_2\gamma$ in Eq. (11):

$$\mathcal{J}(\omega) = \frac{\gamma^2\alpha_1\alpha_2\omega}{\gamma^2\omega^2 + (\alpha_2\gamma - \omega^2)^2}. \quad (\text{B18})$$

Let us now try to find an analytical expression for the Green function in s -space, so that we can compute the analytical expressions for $G(t)$ and $\dot{G}(t)$. For that aim, we need to compute

$$\mathcal{L}\{\chi\}(s) = \frac{2}{\pi} \int_0^\infty d\omega \mathcal{J}(\omega) \frac{\omega}{\omega^2 + s^2}. \quad (\text{B19})$$

For the spectral density in Eq. (B18), the integral we need to solve is

$$\mathcal{L}\{\chi\}(s) = \frac{2\gamma^2\alpha_1\alpha_2}{\pi} \int_0^\infty d\omega \frac{1}{\gamma^2\omega^2 + (\alpha_2\gamma - \omega^2)^2} \frac{\omega^2}{\omega^2 + s^2}, \quad (\text{B20})$$

which we will compute with residue's theorem. For that purpose, let us first symmetrize the integral:

$$\mathcal{L}\{\chi\}(s) = \frac{\gamma^2\alpha_1\alpha_2}{\pi} \int_{-\infty}^\infty d\omega \frac{1}{\gamma^2\omega^2 + (\alpha_2\gamma - \omega^2)^2} \frac{\omega^2}{\omega^2 + s^2} := \int_{-\infty}^\infty d\omega f(\omega) \quad (\text{B21})$$

such that the theorem can be applied in a contour of the complex plane $\mathcal{C}$:

$$\oint_{\mathcal{C}} dz f(z) = 2\pi i \sum_k \text{Res}(f, \omega_k), \quad (\text{B22})$$

where $\{\omega_k\}_k$ are the poles of the function $f(\omega)$. If we close the contour in the upper plane, the poles of $f(\omega)$ can be shown to be

$$\omega_1 = is, \quad (\text{B23})$$

$$\omega_2^\pm = \frac{i\gamma \pm \sqrt{4\alpha_2\gamma - \gamma^2}}{2}, \quad (\text{B24})$$

with corresponding residues

$$\text{Res}(f(\omega), \omega_1) = \frac{is}{2[(\alpha_2\gamma + s^2)^2 - \gamma^2 s^2]}, \quad (\text{B25})$$

$$\text{Res}(f(\omega), \omega_2^+) + \text{Res}(f(\omega), \omega_2^-) = -\frac{i(\alpha_2\gamma + s^2)}{2\gamma[(\alpha_2\gamma + s^2)^2 - \gamma^2 s^2]}. \quad (\text{B26})$$

Therefore, the desired integral will be given by

$$\mathcal{L}\{\chi\}(s) = \frac{\gamma^2 \alpha_1 \alpha_2}{(\alpha_2\gamma + s^2)^2 - \gamma^2 s^2} \left(\alpha_2 + \frac{s^2}{\gamma} - s \right) = \frac{\gamma \alpha_1 \alpha_2}{s^2 + s\gamma + \alpha_2\gamma}, \quad (\text{B27})$$

and the Green function in s -space:

$$\mathcal{G}(s) = \frac{1}{s^2 + \Omega_S^2 + \delta\Omega_S^2 - \frac{\gamma\alpha_1\alpha_2}{s^2 + s\gamma + \alpha_2\gamma}} = \frac{s^2 + s\gamma + \alpha_2\gamma}{(s^2 + \Omega_S^2 + \delta\Omega_S^2)(s^2 + s\gamma + \alpha_2\gamma) - \gamma\alpha_1\alpha_2}. \quad (\text{B28})$$

To solve the dynamics, we need to find $G(t)$ by inverting the Laplace-space Green function. Explicitly,

$$G(t) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} ds e^{st} \mathcal{G}(s), \quad (\text{B29})$$

where $\sigma \in \mathbb{R}$ ensures that the contour path is within the region of convergence of $\mathcal{G}(s)$. In our case, as $\mathcal{G}(s)$ is a rational function in s , i.e., $\mathcal{G}(s) = B(s)/A(s)$, with $B(s) := s^2 + s\gamma + \alpha_2\gamma$ and $A(s) := (s^2 + \Omega_S^2 + \delta\Omega_S^2)(s^2 + s\gamma + \alpha_2\gamma) - \gamma\alpha_1\alpha_2$, the computation can be done analytically. To this end, call $p_1, \dots, p_4$ the roots of $A(s)$, which will also be the poles of $\mathcal{G}(s)$ if no common roots between $A(s)$ and $B(s)$ exist. If we define the residues of $\mathcal{G}(s)$ as $c_i = [(s - p_i)\mathcal{G}(s)]_{s=p_i}$, with $i = 1, \dots, 4$, its inverse Laplace transform is obtained by just summing the residues at its poles,

$$G(t) = (c_1 e^{p_1 t} + \dots + c_4 e^{p_4 t}) \Theta(t). \quad (\text{B30})$$

Here, we have assumed that the poles are simple, which is the case for our choice of $A(s)$ and $B(s)$. Hence, one only needs to compute the poles and residues of $\mathcal{G}(s)$ to construct $G(t)$, which we perform numerically.

b. The covariances

Now, let us compute the symmetrized covariances $\langle x_S^2(t) \rangle$ and $\langle p_S^2(t) \rangle$. For $\langle x_S^2(t) \rangle$, we just need to square Eq. (B14) and take the expectation value:

$$\langle x_S^2(t) \rangle = \langle x_S^2(t) \rangle^{\text{sys}} + \langle x_S^2(t) \rangle^{\text{bath}}, \quad (\text{B31})$$

where the system and bath contributions, $\langle x_S^2(t) \rangle^{\text{sys}}$ and $\langle x_S^2(t) \rangle^{\text{bath}}$, are given by

$$\begin{aligned} \langle x_S^2(t) \rangle^{\text{sys}} &= \dot{G}(t)^2 \langle x_S^2(0) \rangle + G(t)^2 \langle p_S^2(0) \rangle \\ &\quad + G(t) \dot{G}(t) \langle x_S(0) p_S(0) + p_S(0) x_S(0) \rangle, \end{aligned} \quad (\text{B32})$$

$$\langle x_S^2(t) \rangle^{\text{bath}} = \int_0^t d\tau \int_0^t d\tau' G(t-\tau) \nu(\tau-\tau') G(t-\tau'), \quad (\text{B33})$$

where we have assumed an initially thermal bath state, for which $\langle \tilde{B}(\tau) \rangle = 0$. Furthermore, we have defined the symmetrized noise kernel, $\nu(\tau - \tau')$, as

$$\nu(\tau - \tau') := \frac{1}{2} \left\langle \left\{ \tilde{B}(\tau), \tilde{B}(\tau') \right\} \right\rangle, \quad (\text{B34})$$

which can be rewritten in terms of the spectral density as

$$\nu(\tau - \tau') = \frac{1}{\pi} \int_0^\infty d\omega \mathcal{J}(\omega) \coth\left(\frac{\beta\omega}{2}\right) \cos[\omega(\tau - \tau')], \quad (\text{B35})$$

where we have used Eq. (B11). Similarly, squaring Eq. (B17) we get:

$$\begin{aligned} \langle p_S^2(t) \rangle^{\text{sys}} &= \ddot{G}(t)^2 \langle x_S^2(0) \rangle + \dot{G}(t)^2 \langle p_S^2(0) \rangle \\ &\quad + \dot{G}(t)\ddot{G}(t) \langle x_S(0)p_S(0) + p_S(0)x_S(0) \rangle, \end{aligned} \quad (\text{B36})$$

$$\langle p_S^2(t) \rangle^{\text{bath}} = \int_0^t d\tau \int_0^t d\tau' \dot{G}(t-\tau) \nu(\tau-\tau') \dot{G}(t-\tau'). \quad (\text{B37})$$

Now, let us see how to compute Eq. (B35) using again residue's theorem.

c. Noise kernel

First, we rewrite Eq. (B35) in the following suitable way to apply residue's theorem:

$$\nu(t) = \frac{\gamma^2 \alpha_1 \alpha_2}{2\pi} \int_{-\infty}^{\infty} d\omega \operatorname{Re} \left[\frac{\omega}{\gamma^2 \omega^2 + (\alpha_2 \gamma - \omega^2)^2} \coth \left(\frac{\beta \omega}{2} \right) e^{i\omega t} \right] =: \int_{-\infty}^{\infty} d\omega F(\omega). \quad (\text{B38})$$

The function $F(\omega)$ has poles in the following points:

$$\omega_1^{(k)} = i f_k, \quad f_k = \frac{2\pi k}{\beta}, \quad k \in \mathbb{Z} \quad (\text{B39})$$

$$\omega_2^{\pm} = \frac{i\gamma \pm \sqrt{4\alpha_2 \gamma - \gamma^2}}{2}. \quad (\text{B40})$$

Due to the infinite number of poles, a sum over k will appear in the final result, which will be given by

$$\nu(t) = \frac{\gamma \alpha_1 \alpha_2}{2\sqrt{4\alpha_2 \gamma - \gamma^2}} \operatorname{Re} \left[\coth \left(\frac{\beta \omega_2^+}{2} \right) e^{i\omega_2^+ t} - \coth \left(\frac{\beta \omega_2^-}{2} \right) e^{i\omega_2^- t} \right] - \frac{2\gamma^2 \alpha_1 \alpha_2}{\beta} \sum_{k=1}^{\infty} \frac{f_k e^{-f_k t}}{(\alpha_2 \gamma + f_k^2)^2 - \gamma^2 f_k^2}.$$

2. Global GKLS master equation on the system-RC unit

In this section, we derive a global master equation for the system-reaction coordinate unit. Since the system and reaction coordinate may be strongly coupled, we first diagonalize the Hamiltonian of the enlarged system. For the Hamiltonian

$$\hat{H}_{\text{S+RC}} = \frac{1}{2}(p_S^2 + \Omega_1^2 x_S^2) + \frac{1}{2}(P_{\text{RC}}^2 + \Omega_2^2 X_{\text{RC}}^2) - g x_S X_{\text{RC}}, \quad (\text{B41})$$

the normal mode frequencies can be shown to be

$$\Omega_{\pm}^2 = \frac{1}{2} \left(\Omega_1^2 + \Omega_2^2 \pm \sqrt{4g^2 + (\Omega_1^2 - \Omega_2^2)^2} \right), \quad (\text{B42})$$

and the free quadratures:

$$\begin{pmatrix} u_+ \\ u_- \end{pmatrix} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} \begin{pmatrix} x_S \\ X_{\text{RC}} \end{pmatrix}, \quad \text{with} \quad \cos^2(\theta) = \frac{\Omega_1^2 - \Omega_2^2 + \sqrt{4g^2 + (\Omega_1^2 - \Omega_2^2)^2}}{2\sqrt{4g^2 + (\Omega_1^2 - \Omega_2^2)^2}}. \quad (\text{B43})$$

Here, we have defined $\Omega_1^2 := \Omega_S^2 + \delta\Omega_S^2$ and $\Omega_2^2 = \Omega_{RC}^2$. The corresponding 4 jump operators, $\{b_+, b_-, b_+^\dagger, b_-^\dagger\}$ are given by

$$\begin{aligned}
\begin{pmatrix} b_+ \\ b_+^\dagger \\ b_- \\ b_-^\dagger \end{pmatrix} &= \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{\Omega_+} & \frac{i}{\sqrt{\Omega_+}} & 0 & 0 \\ \sqrt{\Omega_+} & \frac{-i}{\sqrt{\Omega_+}} & 0 & 0 \\ 0 & 0 & \sqrt{\Omega_-} & \frac{i}{\sqrt{\Omega_-}} \\ 0 & 0 & \sqrt{\Omega_-} & \frac{-i}{\sqrt{\Omega_-}} \end{pmatrix} \begin{pmatrix} u_+ \\ \dot{u}_+ \\ u_- \\ \dot{u}_- \end{pmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{\Omega_+} & \frac{i}{\sqrt{\Omega_+}} & 0 & 0 \\ \sqrt{\Omega_+} & \frac{-i}{\sqrt{\Omega_+}} & 0 & 0 \\ 0 & 0 & \sqrt{\Omega_-} & \frac{i}{\sqrt{\Omega_-}} \\ 0 & 0 & \sqrt{\Omega_-} & \frac{-i}{\sqrt{\Omega_-}} \end{pmatrix} \begin{pmatrix} \cos(\theta) & 0 & -\sin(\theta) & 0 \\ 0 & \cos(\theta) & 0 & -\sin(\theta) \\ \sin(\theta) & 0 & \cos(\theta) & 0 \\ 0 & \sin(\theta) & 0 & \cos(\theta) \end{pmatrix} \begin{pmatrix} x_S \\ p_S \\ X_{RC} \\ P_{RC} \end{pmatrix} \\
&= \frac{1}{\sqrt{2}} \begin{pmatrix} \sqrt{\Omega_+} \cos(\theta) & \frac{i \cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+} \sin(\theta) & \frac{-i \sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_+} \cos(\theta) & \frac{-i \cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+} \sin(\theta) & \frac{i \sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_-} \sin(\theta) & \frac{i \sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-} \cos(\theta) & \frac{i \cos(\theta)}{\sqrt{\Omega_-}} \\ \sqrt{\Omega_-} \sin(\theta) & \frac{-i \sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-} \cos(\theta) & \frac{-i \cos(\theta)}{\sqrt{\Omega_-}} \end{pmatrix} \begin{pmatrix} x_S \\ p_S \\ X_{RC} \\ P_{RC} \end{pmatrix}. \tag{B44}
\end{aligned}$$

Let us now derive the global master equation for the RC-augmented system. To do that, we first identify some terms in Eq. (13) as:

$$\hat{H} = \hat{H}_{S+RC} - X_{RC} \underbrace{\sum_{k>1} C_k X_k}_{\hat{H}_I} + \frac{1}{2} \underbrace{\sum_{k>1} (P_k^2 + \Omega_k^2 X_k^2)}_{\hat{H}_{RB}}, \tag{B45}$$

so we can write the interacting Hamiltonian in the interaction picture as¹

$$\tilde{H}_I(t) = e^{i(H_{S+RC} + H_{RB})t} H_I e^{-i(H_{S+RC} + H_{RB})t}. \tag{B46}$$

For this Hamiltonian, the dynamics of the system in the interaction picture reads:

$$\dot{\tilde{\rho}} = -i \text{Tr}_B [\tilde{H}_I(t), \tilde{\rho}(0)] - \int_0^t ds \text{Tr}_B [\tilde{H}_I(t), [\tilde{H}_I(s), \tilde{\rho}(s)]]. \tag{B47}$$

As the bath is in a thermal state, the first term vanishes. Hence,

$$\dot{\tilde{\rho}} = - \int_0^t ds \text{Tr}_B [\tilde{H}_I(t), [\tilde{H}_I(s), \tilde{\rho}(s)]] = - \int_0^t ds \text{Tr}_B [\tilde{H}_I(t), [\tilde{H}_I(t-s), \tilde{\rho}_{S+RC}(t-s) \otimes \tilde{\rho}_B]] \tag{B48}$$

$$= - \int_0^t ds \text{Tr}_B [\tilde{H}_I(t) \tilde{H}_I(t-s) \tilde{\rho}_{S+RC}(t-s) \otimes \tilde{\rho}_B - \tilde{H}_I(t-s) \tilde{\rho}_{S+RC}(t-s) \otimes \tilde{\rho}_B \tilde{H}_I(t) + \text{h.c.}] \tag{B49}$$

$$= - \int_0^t ds \left(\tilde{A}(t) \tilde{A}(t-s) \tilde{\rho}_{S+RC}(t-s) - \tilde{A}(t-s) \tilde{\rho}_{S+RC}(t-s) \tilde{A}(t) \right) \text{Tr}_B \left(\tilde{B}(t) \tilde{B}(t-s) \tilde{\rho}_B \right) + \text{h.c.}, \tag{B50}$$

where we changed $s \leftrightarrow t-s$ and defined:

$$A = X_{RC} := \frac{b_- + b_-^\dagger}{\sqrt{2\Omega_-}} \cos \theta - \frac{b_+ + b_+^\dagger}{\sqrt{2\Omega_+}} \sin \theta \Rightarrow \tilde{A}(t) = \frac{b_- e^{-i\Omega_- t} + b_-^\dagger e^{i\Omega_- t}}{\sqrt{2\Omega_-}} \cos \theta - \frac{b_+ e^{-i\Omega_+ t} + b_+^\dagger e^{i\Omega_+ t}}{\sqrt{2\Omega_+}} \sin \theta, \tag{B51}$$

$$B := - \sum_k \frac{C_k}{\sqrt{2\Omega_k}} (b_k + b_k^\dagger) \Rightarrow \tilde{B}(t) = - \sum_k \frac{C_k}{\sqrt{2\Omega_k}} (e^{-i\Omega_k t} b_k + e^{i\Omega_k t} b_k^\dagger). \tag{B52}$$

¹ From here on, we ease the hat notation in Hamiltonians to avoid symbol stacking.

so that

$$\text{Tr}_B \left(\tilde{B}(t) \tilde{B}(t-s) \tilde{\rho}_B \right) = \frac{1}{2} \sum_k \frac{C_k^2}{\Omega_k} \left[e^{-i\Omega_k s} (\mathcal{N}(\Omega_k, T) + 1) + e^{i\Omega_k s} \mathcal{N}(\Omega_k, T) \right], \quad (\text{B53})$$

where $\mathcal{N}(\Omega_k, T)$ is the bosonic occupation number at temperature T and frequency Ω_k . Eq. (B51) shows that the RC coupling operator contains terms oscillating at frequencies $\pm\Omega_{\pm}$. Hence, after the Born–Markov and secular approximations, only terms with matching Bohr frequencies are retained, while the rapidly rotating cross terms between different normal modes are neglected. After these approximations, the master equation is in GKLS form, i.e.

$$\dot{\tilde{\rho}} = \sum_k \left(L_k \tilde{\rho} L_k^\dagger - \frac{1}{2} \left\{ L_k^\dagger L_k, \tilde{\rho} \right\} \right), \quad (\text{B54})$$

with the following set of Lindblad operators:

$$L_1 = \sqrt{\frac{\sin^2 \theta}{\Omega_+} \mathcal{J}_{\text{RC}}(\Omega_+) (\mathcal{N}(\Omega_+, T) + 1)} b_+, \quad (\text{B55})$$

$$L_2 = \sqrt{\frac{\sin^2 \theta}{\Omega_+} \mathcal{J}_{\text{RC}}(\Omega_+) \mathcal{N}(\Omega_+, T)} b_+^\dagger, \quad (\text{B56})$$

$$L_3 = \sqrt{\frac{\cos^2 \theta}{\Omega_-} \mathcal{J}_{\text{RC}}(\Omega_-) (\mathcal{N}(\Omega_-, T) + 1)} b_-, \quad (\text{B57})$$

$$L_4 = \sqrt{\frac{\cos^2 \theta}{\Omega_-} \mathcal{J}_{\text{RC}}(\Omega_-) \mathcal{N}(\Omega_-, T)} b_-^\dagger, \quad (\text{B58})$$

where we have neglected the imaginary Lamb shift terms. In parallel, it is known that for a Lindbladian master equation of the form of Eq. (B54), the equations of motion for the first and second moments are given by Eqs. (4). There, the matrices A and D are defined as follows:

$$A := \Omega [G + \text{Im}(C^\dagger C)] \quad (\text{B59})$$

$$D := \Omega \text{Re}(C^\dagger C) \Omega^T \quad (\text{B60})$$

with here I changed $\{\Omega_S, \Omega_{\text{RC}}\} \mapsto \{\Omega_1, \Omega_2\}$. check it again

$$G := \begin{pmatrix} \Omega_1^2 & 0 & -g & 0 \\ 0 & 1 & 0 & 0 \\ -g & 0 & \Omega_2^2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad C := (c_1^T; c_2^T; c_3^T; c_4^T)^T, \quad (\text{B61})$$

where c_k is defined such that the Lindblad operators are given by $L_k = c_k^T R$. some comment on dropping the renormalization Using the Lindblad operators in Eqs. (B55)-(B58), one can build the matrix C by multiplying by the corresponding weights:

$$C = \begin{pmatrix} \frac{\sqrt{\mathcal{J}_{\text{RC}}(\Omega_+) [\mathcal{N}(\Omega_+, T) + 1]}}{\sqrt{2\Omega_+}} \sin \theta \\ \frac{\sqrt{\mathcal{J}_{\text{RC}}(\Omega_+) \mathcal{N}(\Omega_+, T)}}{\sqrt{2\Omega_+}} \sin \theta \\ \frac{\sqrt{\mathcal{J}_{\text{RC}}(\Omega_-) [\mathcal{N}(\Omega_-, T) + 1]}}{\sqrt{2\Omega_-}} \cos \theta \\ \frac{\sqrt{\mathcal{J}_{\text{RC}}(\Omega_-) \mathcal{N}(\Omega_-, T)}}{\sqrt{2\Omega_-}} \cos \theta \end{pmatrix} \bullet \begin{pmatrix} \sqrt{\Omega_+} \cos(\theta) & \frac{i \cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+} \sin(\theta) & \frac{-i \sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_+} \cos(\theta) & \frac{-i \cos(\theta)}{\sqrt{\Omega_+}} & -\sqrt{\Omega_+} \sin(\theta) & \frac{i \sin(\theta)}{\sqrt{\Omega_+}} \\ \sqrt{\Omega_-} \sin(\theta) & \frac{i \sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-} \cos(\theta) & \frac{i \cos(\theta)}{\sqrt{\Omega_-}} \\ \sqrt{\Omega_-} \sin(\theta) & \frac{-i \sin(\theta)}{\sqrt{\Omega_-}} & \sqrt{\Omega_-} \cos(\theta) & \frac{-i \cos(\theta)}{\sqrt{\Omega_-}} \end{pmatrix}, \quad (\text{B62})$$

where $A \bullet B := \text{diag}(A)B$ represents element-wise multiplication. In this case, the D matrix is given by:

$$D = \begin{pmatrix} d_{11} & 0 & d_{13} & 0 \\ 0 & d_{22} & 0 & d_{24} \\ d_{31} & 0 & d_{33} & 0 \\ 0 & d_{42} & 0 & d_{44} \end{pmatrix} \quad (\text{B63})$$

with

$$d_{11} = \frac{\sin^2(2\theta)}{8\Omega_+^2\Omega_-^2} \left[\mathcal{J}_{\text{RC}}(\Omega_-)\Omega_+^2(2\mathcal{N}(\Omega_-, T) + 1) + \mathcal{J}_{\text{RC}}(\Omega_+)\Omega_-^2(2\mathcal{N}(\Omega_+, T) + 1) \right] \quad (\text{B64})$$

$$d_{22} = \frac{\sin^2(2\theta)}{8} \left[\mathcal{J}_{\text{RC}}(\Omega_-)(2\mathcal{N}(\Omega_-, T) + 1) + \mathcal{J}_{\text{RC}}(\Omega_+)(2\mathcal{N}(\Omega_+, T) + 1) \right] \quad (\text{B65})$$

$$d_{33} = \frac{\mathcal{J}_{\text{RC}}(\Omega_-)\Omega_+^2(2\mathcal{N}(\Omega_-, T) + 1)\cos^4\theta + \mathcal{J}_{\text{RC}}(\Omega_+)\Omega_-^2(2\mathcal{N}(\Omega_+, T) + 1)\sin^4\theta}{2\Omega_+^2\Omega_-^2} \quad (\text{B66})$$

$$d_{44} = \frac{1}{2} \left[\mathcal{J}_{\text{RC}}(\Omega_-)(2\mathcal{N}(\Omega_-, T) + 1)\cos^4\theta + \mathcal{J}_{\text{RC}}(\Omega_+)(2\mathcal{N}(\Omega_+, T) + 1)\sin^4\theta \right] \quad (\text{B67})$$

$$d_{24} = d_{42} = \frac{\sin(2\theta)}{4} \left[\mathcal{J}_{\text{RC}}(\Omega_-)(2\mathcal{N}(\Omega_-, T) + 1)\cos^2\theta - \mathcal{J}_{\text{RC}}(\Omega_+)(2\mathcal{N}(\Omega_+, T) + 1)\sin^2\theta \right], \quad (\text{B68})$$

$$d_{13} = d_{31} = \frac{\sin(2\theta)}{4\Omega_+^2\Omega_-^2} \left[\mathcal{J}_{\text{RC}}(\Omega_-)\Omega_+^2(2\mathcal{N}(\Omega_-, T) + 1)\cos^2\theta - \mathcal{J}_{\text{RC}}(\Omega_+)\Omega_-^2(2\mathcal{N}(\Omega_+, T) + 1)\sin^2\theta \right]. \quad (\text{B69})$$

For the A matrix we obtain:

$$A = \begin{pmatrix} a_{11} & 1 & a_{13} & 0 \\ -\Omega_1^2 & a_{22} & g & a_{24} \\ a_{31} & 0 & a_{33} & 1 \\ g & a_{42} & -\Omega_2^2 & a_{44} \end{pmatrix} \quad (\text{B70})$$

where

$$a_{11} = a_{22} = -\frac{\sin^2(2\theta)}{8\Omega_+\Omega_-} \left(\mathcal{J}_{\text{RC}}(\Omega_-)\Omega_+ + \mathcal{J}_{\text{RC}}(\Omega_+)\Omega_- \right), \quad (\text{B71})$$

$$a_{33} = a_{44} = -\frac{1}{2\Omega_+\Omega_-} \left[\mathcal{J}_{\text{RC}}(\Omega_-)\Omega_+\cos^4\theta + \mathcal{J}_{\text{RC}}(\Omega_+)\Omega_-\sin^4\theta \right], \quad (\text{B72})$$

$$a_{13} = a_{31} = a_{24} = a_{42} = -\frac{\sin(2\theta)}{4\Omega_+\Omega_-} \left(\mathcal{J}_{\text{RC}}(\Omega_-)\Omega_+ - \mathcal{J}_{\text{RC}}(\Omega_+)\Omega_- \right) \cos^2\theta + \frac{\sin(2\theta)}{4\Omega_+\Omega_-} \mathcal{J}_{\text{RC}}(\Omega_+)\Omega_-. \quad (\text{B73})$$

To see that the signs are indeed correct, we take the limit of zero dissipation and study the equations of motion. That is, if $\mathcal{J}_{\text{RC}}(\Omega_{\pm}) = 0$. This gives the following A matrix:

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ -\Omega_1^2 & 0 & g & 0 \\ 0 & 0 & 0 & 1 \\ g & 0 & -\Omega_2^2 & 0 \end{pmatrix}, \quad (\text{B74})$$

which gives the following equations of motion:

$$\dot{x}_{\text{S}} = p_{\text{S}}, \quad (\text{B75})$$

$$\dot{p}_{\text{S}} = -\Omega_1^2 x_{\text{S}} + g X_{\text{RC}}, \quad (\text{B76})$$

$$\dot{X}_{\text{RC}} = P_{\text{RC}}, \quad (\text{B77})$$

$$\dot{P}_{\text{RC}} = g x_{\text{S}} - \Omega_2^2 X_{\text{RC}}, \quad (\text{B78})$$

which coincide with the Hamilton equations for the Hamiltonian $\hat{H}_{\text{S+RC}}$.

3. Local GKLS master equation on the system

Now, we solve the GKLS master equation for the original system-bath Hamiltonian in Eq. (8). As we anticipated in the main text, this approach will fail in the strong coupling regime, making the global GKLS master equation with RC—see Appendix B 2—better in such regime.

To solve the dynamics, we again use Eqs. (4), such that the dynamics can be fully described by finding the matrices A and D . We take the Lindblad operators to be [7, 8]

$$L_1 = \sqrt{\kappa(\mathcal{N}(\Omega_S, T) + 1)}a, \quad (\text{B79})$$

$$L_2 = \sqrt{\kappa\mathcal{N}(\Omega_S, T)}a^\dagger, \quad (\text{B80})$$

with $\kappa := \mathcal{J}(\Omega_S)/\Omega_S$ in this case. The matrix C is then given by

$$C = \begin{pmatrix} \sqrt{\frac{\kappa\Omega_S(\mathcal{N}(\Omega_S, T) + 1)}{2}} & \sqrt{\frac{\kappa\Omega_S\mathcal{N}(\Omega_S, T)}{2}} \\ i\sqrt{\frac{\kappa\mathcal{N}(\Omega_S, T)}{2\Omega_S}} & -i\sqrt{\frac{\kappa\mathcal{N}(\Omega_S, T)}{2\Omega_S}} \end{pmatrix}. \quad (\text{B81})$$

Hence, the drift and diffusion matrices can be shown to be

$$A = \begin{pmatrix} -\frac{\kappa}{2} & 1 \\ -\Omega_S^2 & -\frac{\kappa}{2} \end{pmatrix}, \quad D = \begin{pmatrix} \frac{\kappa}{\Omega_S} \left(\mathcal{N}(\Omega_S, T) + \frac{1}{2} \right) & 0 \\ 0 & \kappa\Omega_S \left(\mathcal{N}(\Omega_S, T) + \frac{1}{2} \right) \end{pmatrix}. \quad (\text{B82})$$

Given these matrices, we fully simulate the dynamics of the system via the evolution of its first and second moments in Eq. (4).

4. Comparison between the different approaches

In this section, we numerically solve the exact covariances in Eqs. (B14) and (B17) and compare them with the solution given by the GKLS global master equation with the reaction coordinate, see Sec. B2, as well as a local GKLS master equation on the bare Hamiltonian, see Sec. B3, to showcase the strength of the reaction coordinate approach within the strong coupling regime. In Fig. 2 we compare the dynamics for the three approaches, both in the early-time

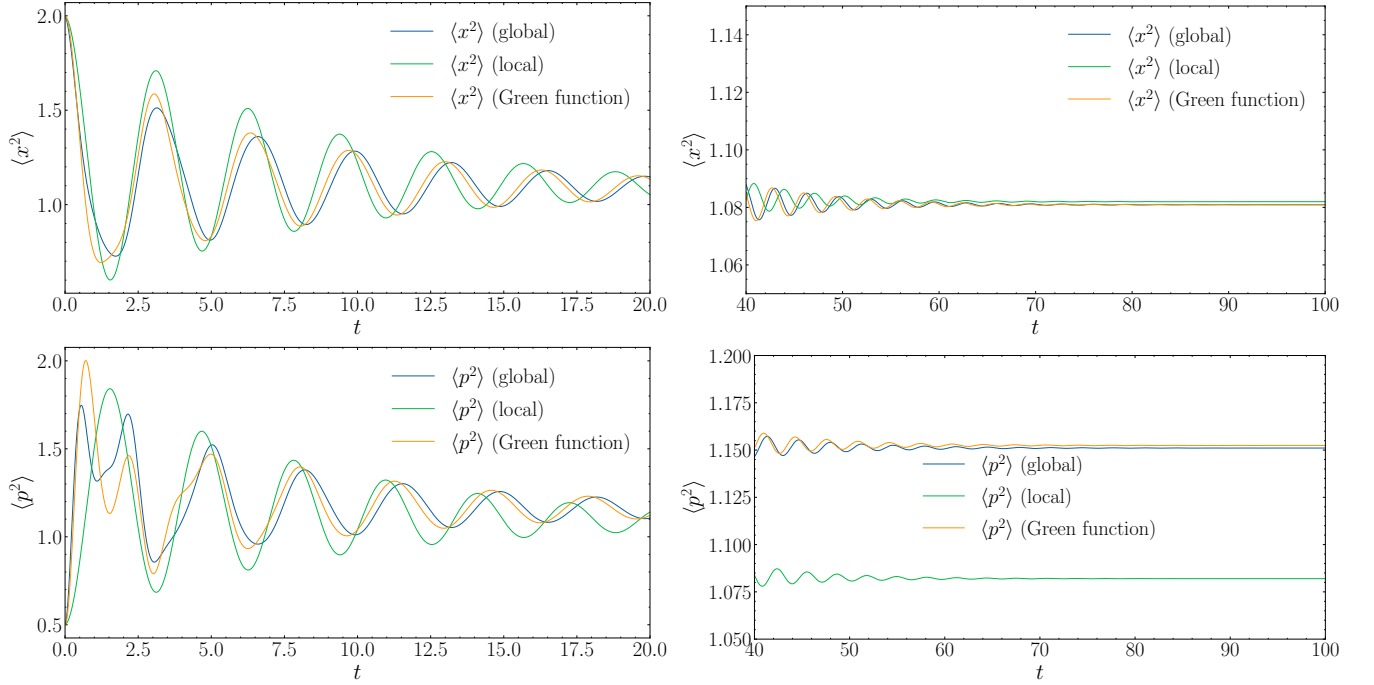


Figure 2. $\Omega_S = 1.0, \alpha_1 = 1, \alpha_2 = 10, \gamma = 1, T = 1$.

regime and in the steady state case. As we see,

Appendix C: The FI of the homodyne current

Take a Gaussian system characterized by $\{d(\theta; \vec{W}_{m-1}), \sigma(\theta; \vec{W}_{m-1})\}$, with $\vec{W}_k = \{W_k, \dots, W_1\}$ being the sequence of measurement outcome history. That is, the system is monitored continuously, getting an outcome W_m at step m with the conditional probability $p(W_m|\vec{W}_{m-1}; \theta) = \mathcal{N}(\xi(\theta; \vec{W}_{m-1}), V(\theta; \vec{W}_{m-1}))$, where $\xi(\theta; \vec{W}_{m-1}) = \sqrt{\lambda\delta t}Ld(\theta; \vec{W}_{m-1})$ and $V(\theta; \vec{W}_{m-1}) = \sigma_{\text{env}} + \sigma^M + \lambda\delta t(\sigma(\theta; \vec{W}_{m-1}) - \sigma_{\text{env}})$. Here, $L = \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \end{bmatrix}$, captures the system displacement from the displacement of the S-RC, that is $d_S(\theta; W) = Ld(\theta; W)$. Let's restrict to the case where we ignore the $\lambda\delta t$ term in the variance, and hence we can replace $V = V(\theta; \vec{W}_{m-1})$. Furthermore, we consider a scenario in which the system has reached a stationary state, that is, its covariance matrix does not evolve. Conditioned on this outcome, we will update the state to

$$d(\theta; \vec{W}_{m-1}) \mapsto (I + \delta t A(\theta))d(\theta; \vec{W}_{m-1}) + \sqrt{\lambda}K(\theta)Y(\theta; \vec{W}_m) = d(\theta; \vec{W}_m) \quad \text{with} \quad K(\theta) = \begin{bmatrix} \sigma_S(\theta) - \sigma_{\text{env}} \\ C^T(\theta) \end{bmatrix} V^{-1}$$

$$\sigma(\theta) \mapsto \sigma(\theta), \quad \text{which is the solution to} \quad A(\theta)\sigma(\theta) + \sigma(\theta)A^T(\theta) + D(\theta) - \lambda K(\theta)VK^T(\theta) = 0. \quad (\text{C1})$$

That is, we are working in a regime where the covariance matrix has reached its steady state. Here, we have defined $Y(\theta; \vec{W}_m) = \sqrt{\delta t}W_m - \delta t\sqrt{\lambda}d_S(\theta; \vec{W}_{m-1})$ which is a Wiener process, with vanishing mean and a variance that is proportional to δtV . Note that the observed data is W_m and not $Y(\theta; \vec{W}_m)$. To obtain the FI of the entire data, we note the following. The data itself is not necessarily independent. For instance, we have $p(\vec{W}_n; \theta) = p(W_n|\vec{W}_{n-1}; \theta)p(\vec{W}_{n-1}; \theta) = \Pi_{k=1}^n p(W_k|\vec{W}_{k-1}; \theta)$, where $\vec{W}_0$ is no data, i.e., $p(W_1|\vec{W}_0; \theta) \equiv p(W_1; \theta)$. Then, the FI can be cast as

$$\begin{aligned} F_{\text{total}}(\theta) &= \left\langle [\partial_\theta \log p(\vec{W}_n; \theta)]^2 \right\rangle = \int d\vec{W}_n p(\vec{W}_n; \theta) \left[\sum_{m=1}^n \partial_\theta \log p(W_m|\vec{W}_{m-1}; \theta) \right]^2 \\ &= \sum_{m=1}^n \int d\vec{W}_n p(\vec{W}_n; \theta) \left[\partial_\theta \log p(W_m|\vec{W}_{m-1}; \theta) \right]^2 \quad \text{since the cross terms disappear} \\ &= \sum_{m=1}^n \int d\vec{W}_m p(\vec{W}_m; \theta) \left[\partial_\theta \log p(W_m|\vec{W}_{m-1}; \theta) \right]^2 \quad \text{since we can integrate the events after } W_m \text{ to get one} \\ &= \sum_{m=1}^n \int d\vec{W}_{m-1} p(\vec{W}_{m-1}; \theta) \int dW_m p(W_m|\vec{W}_{m-1}; \theta) \left[\partial_\theta \log p(W_m|\vec{W}_{m-1}; \theta) \right]^2 \\ &= \sum_{m=1}^n \int d\vec{W}_{m-1} p(\vec{W}_{m-1}; \theta) F_m(\vec{W}_{m-1}; \theta) \\ &= \sum_{m=1}^n F_m(\theta), \quad \text{with the average FI gain} \quad F_m(\theta) := \int d\vec{W}_{m-1} p(\vec{W}_{m-1}; \theta) F_m(\vec{W}_{m-1}; \theta), \\ &\quad \text{and with the conditional FI} \quad F_m(\vec{W}_{m-1}; \theta) := \int dW_m p(W_m|\vec{W}_{m-1}; \theta) \left[\partial_\theta \log p(W_m|\vec{W}_{m-1}; \theta) \right]^2. \end{aligned} \quad (\text{C2})$$

Since we are dealing with Gaussian distributions (where the variance is independent of the parameter, but the mean depends on it), the average FI gain from the m th observation can be alternatively written as

$$\begin{aligned} F_m(\theta) &= (\lambda\delta t) \mathbb{E} \left[2\partial_\theta d(\vec{W}_m; \theta)^T L^T V^{-1} L \partial_\theta d(\vec{W}_m; \theta) \right]_{\vec{W}_m} = 2(\lambda\delta t) \text{Tr} \left[\mathbb{E} \left[\partial_\theta d(\vec{W}_m; \theta)^T L^T V^{-1} L \partial_\theta d(\vec{W}_m; \theta) \right]_{\vec{W}_m} \right] \\ &= 2(\lambda\delta t) \mathbb{E} \left[\text{Tr} \left[\partial_\theta d(\vec{W}_m; \theta)^T L^T V^{-1} L \partial_\theta d(\vec{W}_m; \theta) \right] \right]_{\vec{W}_m} = 2(\lambda\delta t) \mathbb{E} \left[\text{Tr} \left[L^T V^{-1} L \partial_\theta d(\vec{W}_m; \theta) \partial_\theta d(\vec{W}_m; \theta)^T \right] \right]_{\vec{W}_m} \\ &= 2(\lambda\delta t) \text{Tr} \left[L^T V^{-1} L \mathbb{E} \left[\partial_\theta d(\vec{W}_m; \theta) \partial_\theta d(\vec{W}_m; \theta)^T \right]_{\vec{W}_m} \right] \end{aligned} \quad (\text{C3})$$

Summing up over $F_m(\theta)$ we can find the total FI

$$F_{\text{total}}(\theta) = \sum_{m=1}^n F_m(\theta) = 2\lambda\delta t \sum_{m=1}^n \text{Tr} \left[L^T V^{-1} L \mathbb{E} \left[\partial_\theta d(\vec{W}_m; \theta) \partial_\theta d(\vec{W}_m; \theta)^T \right]_{\vec{W}_m} \right]. \quad (\text{C4})$$

Since $\delta t \ll 1$ we can change the summations with integrals. By defining $\tau = n\delta t$ and $f(t = m\delta t) = \mathbb{E} \left[\partial_\theta d(\vec{W}_m; \theta) \partial_\theta d(\vec{W}_m; \theta)^T \right]_{\vec{W}_m}$ we have

$$F_{\text{total}}(\theta) = 2\lambda \int_0^\tau dt \text{Tr}[L^T V^{-1} L f(t)] = 2\lambda \text{Tr} \left[L^T V^{-1} L \int_0^\tau dt f(t) \right]. \quad (\text{C5})$$

We need to evaluate the integral, and we are interested in large data, i.e., $\tau \gg 1$. To proceed further, note that we have

$$\begin{aligned} \delta d(\vec{W}_m; \theta) &= d(\vec{W}_m; \theta) - d(\vec{W}_{m-1}; \theta) \\ &= \delta t A d(\vec{W}_{m-1}; \theta) + \sqrt{\lambda} K Y(\vec{W}_m; \theta), \\ \delta \partial_\theta d(\vec{W}_m; \theta) &= \partial_\theta d(\vec{W}_m; \theta) - \partial_\theta d(\vec{W}_{m-1}; \theta) \\ &= \delta t [A - \lambda K L] \partial_\theta d(\vec{W}_{m-1}; \theta) + \delta t A' d(\vec{W}_{m-1}; \theta) + \sqrt{\lambda} K' Y(\vec{W}_m; \theta). \end{aligned} \quad (\text{C6})$$

We will be interested in calculating $f(t)$ for the FI. This can be evaluated by using the augmented system $Z(\vec{W}_m; \theta) = \begin{bmatrix} d(\vec{W}_m; \theta) \\ \partial_\theta d(\vec{W}_m; \theta) \end{bmatrix}$, such that

$$\delta Z(\vec{W}_m; \theta) = \delta t \mathcal{A} Z(\vec{W}_{m-1}; \theta) + \mathcal{K} Y(\vec{W}_m; \theta), \quad \text{with } \mathcal{A} = \begin{bmatrix} A & 0 \\ A' & A - \lambda K L \end{bmatrix}, \quad \text{and } \mathcal{K} = \begin{bmatrix} \sqrt{\lambda} K \\ \sqrt{\lambda} K' \end{bmatrix}. \quad (\text{C7})$$

Then, by defining $\Sigma(\theta) = \mathbb{E}[Z(\vec{W}_m; \theta) Z^T(\vec{W}_m; \theta)]_{W_m}$ to the leading order in δt we have

$$\delta \Sigma(\theta) = \delta t (\mathcal{A} \Sigma(\theta) + \Sigma(\theta) \mathcal{A}^T + \mathcal{K} V \mathcal{K}^T), \quad (\text{C8})$$

where we used that $Y(\vec{W}_m; \theta)$ are iid with vanishing mean and variance equal to $\delta t V$. The steady state is the solution to the Lyapunov equation

$$\mathcal{A} \Sigma^{\text{ss}}(\theta) + \Sigma(\theta)^{\text{ss}} \mathcal{A}^T + \mathcal{K} V \mathcal{K}^T = 0. \quad (\text{C9})$$

Then, at the steady state, we have $f(t \gg 1) = \mathcal{Q} \Sigma^{\text{ss}}(\theta) \mathcal{Q}^T$, with $\mathcal{Q} = [0, \quad I]$ helping to extract the bottom right block of $\Sigma^{\text{ss}}(\theta)$. Then for large signals, we can approximate the FI as follows

$$F_{\text{total}}(\theta) = 2\tau \lambda \text{Tr}[L^T V^{-1} L \mathcal{Q} \Sigma^{\text{ss}}(\theta) \mathcal{Q}^T]. \quad (\text{C10})$$

1. Bayesian Cramer Rao bound

The FI that we obtained can be used to obtain a BCRB. The relative error can be cast as

$$\text{RMSE} = \int d\theta p(\theta) \int d\vec{W}_n p(\vec{W}_n | \theta) \frac{[\hat{\theta}(\vec{W}_n) - \theta]^2}{\theta^2}. \quad (\text{C11})$$

Van Trees inequality is originally written for the MSE. However, in thermometry we deal with RMSE or MSLE. By using the Van Trees inequality we have

$$\begin{aligned} \text{RMSE}^{-1} &\leq Q[\theta] + \int d\theta \theta^2 d\vec{W}_n p(\theta, \vec{W}) [\partial_\theta \log p(\vec{W}_n | \theta)]^2 = Q[\theta] + \int d\theta p(\theta) \theta^2 \int d\vec{W}_n p(\vec{W}_n | \theta) [\partial_\theta \log p(\vec{W}_n | \theta)]^2 \\ &= Q[\theta] + \int d\theta p(\theta) \theta^2 F_{\text{total}}(\theta), \end{aligned} \quad (\text{C12})$$

with the prior information

$$Q[\theta] = \int d\theta p(\theta) \theta^2 [\partial_\theta \log(p(\theta))]^2 \quad (\text{C13})$$